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United States Patent Application Publication

20250285642

Kind Code

A1

Publication Date

September 11, 2025

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Spin Injection Assisted Magnetic Recording

Abstract

A spin injection assisted magnetic recording structure is disclosed wherein a ferromagnetic (FM) layer and a spin preservation (SP) layer are formed between a main pole (MP) trailing side and a write shield (WS). Current (Ia) is injected into the FM layer and flows through the SP layer to the WS. As a result, spin polarized electrons from the FM layer generate a magnetization that enhances a local WS magnetization and return field. A lead to the FM layer may be stitched to enable lower resistance and improve reliability. The FM layer is separated from the MP trailing side with a write gap, and is recessed from the ABS to allow more overlap with the SP layer for lower current density while maintaining performance. Higher linear density and area density capability, and better reliability are achieved.

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Family ID: 1000008618369

Appl. No.: 19/208182

Filed: May 14, 2025

Related U.S. Application Data

parent US division 18359436 20230726 parent-grant-document US 12322419 child US 19208182
parent US division 17534574 20211124 parent-grant-document US 11756577 child US 18359436
parent US division 16563198 20190906 parent-grant-document US 11189304 child US 17534574

Publication Classification

Int. Cl.: G11B5/012 (20060101); G11B5/02 (20060101); G11B5/11 (20060101); G11B5/127 (20060101); G11B5/31 (20060101)

U.S. Cl.:

CPC **G11B5/012** (20130101); **G11B5/02** (20130101); **G11B5/11** (20130101); **G11B5/1278** (20130101); **G11B5/3133** (20130101); **G11B5/3146** (20130101); **G11B5/315** (20130101);

Background/Summary

[0001] This is a divisional application of U.S. patent application Ser. No. 18/359,436 filed on Jul. 26, 2023, which is a divisional application of U.S. patent application Ser. No. 17/534,574 filed on Nov. 24, 2021 and issued as U.S. Pat. No. 11,756,577, which is a divisional application of U.S. patent application Ser. No. 16/563,198; filed on Sep. 6, 2019, and issued as U.S. Pat. No. 11,189,304, assigned to a common assignee, and herein incorporated by reference in their entirety.

RELATED PATENT APPLICATIONS

[0002] This application is related to the following: U.S. Pat. Nos. 10,325,618; and 10,424,326; assigned to a common assignee, and herein incorporated by reference in their entirety.

TECHNICAL FIELD

[0003] The present disclosure relates to a design for a spin injection assisted magnetic recording structure wherein a ferromagnetic (FM) layer adjoins one or two spin preserving (SP) layers in a write gap (WG), and a current is injected across the FM layer and at least one SP layer, and into one or both of a main pole (MP) and write shield (WS) to improve a local magnetic field and field gradient at one or both of a MP/WG interface and a WG/WS interface, respectively, thereby increasing linear density capability and area density capability (ADC) for perpendicular magnetic recording (PMR) applications.

BACKGROUND

[0004] As the data areal density in hard disk drive (HDD) writing increases, write heads and media bits are both required to be made in smaller sizes. However, as the write head size shrinks, its writability degrades. In particular, perpendicular magnetic recording (PMR) performance is limited by the saturation magnetization (M_s) of available magnetic materials. To improve writability, new technology is being developed that assists writing to a media bit. Two main approaches currently being investigated are thermally assisted magnetic recording (TAMR) and microwave assisted magnetic recording (MAMR). The latter is described by J-G. Zhu et al. in “Microwave Assisted Magnetic Recording”, IEEE Trans. Magn., vol. 44, pp. 125-131 (2008). MAMR uses a spin torque device to generate a high frequency field that reduces the coercive field of a medium bit thereby allowing the bit to be switched with a lower main pole field. A third approach called STRAMR (spin torque reversal assisted magnetic recording) relies on spin torque to reverse a magnetization in a flux generating layer (FGL) in the write gap to increase reluctance and force more magnetic flux from the MP at the ABS. STRAMR is described in U.S. Pat. No. 6,785,092.

[0005] Spin transfer (spin torque) devices are based on a spin-transfer effect that arises from the spin dependent electron transport properties of ferromagnetic-spacer-ferromagnetic multilayers. When a spin-polarized current passes through a FM1/NM/FM2 multilayer in a CPP (current perpendicular to plane) configuration where FM1 and FM2 are first and second FM layers and NM is a non-magnetic spacer, the spin angular momentum of electrons from FM1 that is incident on FM2 interacts with magnetic moments of FM2 near the NM/FM2 interface. Through this interaction, the electrons transfer a portion of their angular momentum to FM2. As a result, spin-polarized current can switch the FM2 magnetization direction or enhance FM2 magnetization depending on the current density. Spin transfer devices are also known as spintronic devices and may have ferromagnetic (FM) layers with a perpendicular magnetic anisotropy (PMA) component where

magnetization is aligned substantially perpendicular to the plane of the FM layer. These devices have an advantage over devices based on in-plane anisotropy in that they can satisfy the thermal stability requirement but also have less limits on cell aspect ratio. As a result, spintronic structures based on PMA are capable of scaling for higher packing density, which is a key challenge for future MRAM (Magnetoresistive Random Access Memory) and spin torque transfer (STT)-MRAM applications, and for other spintronic devices such as microwave generators and assist structures for PMR.

[0006] Related U.S. Pat. No. 10,424,326 discloses a magnetic flux guiding device that is a STRAMR approach. STRAMR devices typically require a high current density to flip a magnetization in a FM layer in the WG, and this issue is substantially overcome by placing spin polarization (SP) layers on each side of the FM layer so that an additive spin torque on the FM layer magnetization is generated to allow a reduced current density. However, a new assist design is desired that does not rely on spin flipping to enhance the write field, and has flexibility in increasing one or both of a local magnetic field and field gradient at the main pole/WG interface and at the WG/write shield interface thereby improving linear density and areal density capability.

SUMMARY

[0007] One objective of the present disclosure is to provide a PMR writer wherein a spin injection assisted magnetic recording (SIAMR) device is formed in a write gap (WG) and enables spin polarized electrons to flow into one or both of a write shield at a SIAMR/WS interface and into a MP at a MP/SIAMR interface thereby enhancing the return field and write field, respectively, to improve linear density and ADC.

[0008] A second objective of the present disclosure is to provide a SIAMR device according to the first objective that also provides improved reliability compared with conventional PMR writers with MAMR capability.

[0009] According to one embodiment of the present invention, these objectives are achieved with a PMR writer design wherein a SIAMR device comprises a FM layer and at least one adjoining spin preserving (SP) layer in the WG between a MP and a WS. In a first embodiment, the SIAMR device further includes a pair of so-called spin killing layers made of Ta, W, Pt, Ru, Ti, or Ir on a side of the FM layer that is opposite to the FM side that adjoins the SP layer. The SP layer contacts the WS, and applied current flows from the MP to WS so that spin polarized electrons from the FM layer provide a magnetization that enhances a local WS field proximate to the SP/WS interface and a return field in the WS. In an alternative embodiment, when the SP layer contacts the MP and applied current ($I_{sub.a}$) flows from the WS through the FM layer and SP layer into the MP, $I_{sub.a}$ produces a magnetization that enhances a local MP field proximate to the MP/SP interface, and the write field. There are leads from the MP and WS that are connected to a direct current (dc) source to enable $I_{sub.a}$ to be applied across the SIAMR device during a write process.

[0010] According to a second embodiment, the features of the first embodiment are retained except the spin killing layers are replaced with a dielectric (WG) layer, and $I_{sub.a}$ is applied through a lead to the FM layer and across the SP layer to the WS. In both of the first and second embodiments, each of the FM and SP layers has a front side at the air bearing surface (ABS), and the SP layer has a backside at a WS throat height (TH). A backside of the FM layer is typically at a height substantially greater than the WS TH in the second embodiment. The FM layer may be a single layer or multilayer that is comprised of one or more of Fe, Co, CoFe, NiFe, CoFeNi, and alloys thereof such as CoB, FeB, CoFeB, and CoFeNiB, or alloys with one or more of Ta, Zr, and Cr such as CoTaZr. The SP layer is a material comprised of one or more of Cu, Au, Ag, Ru, Cr, and Al while the dielectric (WG) layer may be an oxide or nitride of Al, Mg, Si, Ti, Ta, Hf, or Zr.

[0011] A third embodiment retains all the features of the second embodiment except the backside of the SP layer is extended to a height greater than the WS TH to enable greater current density in the spin polarized current through the SP layer.

[0012] According to a fourth embodiment that is a modification of the third embodiment, a front

side of the FM layer is recessed to the WS TH, for example, to allow lower current density between the FM and SP layers because greater overlap of the two layers is possible in the cross-track direction outside the tight confines of the WG proximate to the ABS. Therefore, improved reliability is expected since lower current density means lower risk of electromigration of metals or alloys within the FM and SP layers.

[0013] In a fifth embodiment, the dielectric layer adjoining one side of the FM layer in the fourth embodiment is replaced with a first SP layer (SP1) such that spin polarized current from the FM layer flows through SP1 to the MP and through a second SP layer (SP2) to the WS to enhance localized MP and localized WS magnetization, respectively, thereby improving the write field and return field. Optionally, one or both of the FM layer front side is at the ABS, and the SP1 and SP2 backsides are at the WS TH.

[0014] According to a sixth embodiment, any of the second through fifth embodiments is modified to use a stitched lead between the dc source and the FM layer. Thus, a back portion of the FM layer may be stitched to a SP material or to a spin killing material. As a result, the lead resistance is lowered, which in turn reduces operating temperature to improve reliability.

[0015] In the seventh embodiment, the FM layer in the second embodiment is omitted and the SP layer adjoins both of the WS and MP so that when I_{a} is applied from the MP to the WS, a portion of the MP proximate to the MP/SP interface serves as the FM layer, which spin polarizes $I_{\text{sub.a}}$. A first portion of the SP layer proximate to the MP trailing side is preferably recessed beyond the WS TH to avoid reducing MP magnetization proximate to the ABS, and a second portion of the SP layer adjoining the WS has a front side at the ABS, and a backside at the same height as the first portion.

[0016] A process sequence is also provided for forming a SIAMR stack of layers according to an embodiment of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a perspective view of a head arm assembly of the present disclosure.

[0018] FIG. 2 is side view of a head stack assembly of the present disclosure.

[0019] FIG. 3 is a plan view of a magnetic recording apparatus of the present disclosure.

[0020] FIG. 4 is a down-track cross-sectional view of a combined read-write head with leading and trailing loop pathways for magnetic flux return to the main pole in the PMR writer according to an embodiment of the present disclosure.

[0021] FIG. 5 is an ABS view of a SIAMR device according to a first embodiment of the present disclosure wherein a stack comprised of two spin killing layers, a FM layer, and a SP layer are formed on a MP trailing side and in a WG such that the SP layer adjoins a WS bottom surface.

[0022] FIG. 6A is a down-track cross-sectional view of the SIAMR device in FIG. 5 where a current $I_{\text{sub.a}}$ is applied from the MP and across the FM and SP layers to enhance a local WS magnetic field and WS return field.

[0023] FIG. 6B is an alternative first embodiment where the stacking order of SIAMR device layers in FIG. 6A is reversed, and $I_{\text{sub.a}}$ is applied from the TS and across the FM and SP layers to enhance a local MP magnetic field and write field.

[0024] FIG. 7 is a down-track cross-sectional view where the spin killing layers in FIG. 6A are replaced with a dielectric layer and $I_{\text{sub.a}}$ is injected into the FM layer and flows to the WS according to a second embodiment of the present disclosure.

[0025] FIG. 8 is a down-track cross-sectional view depicting a third embodiment of the present disclosure wherein a backside of the SP layer in FIG. 7 is extended beyond a WS throat height (TH), and $I_{\text{sub.a}}$ flows from the FM layer to WS.

[0026] FIG. **9** is a down-track cross-sectional view of a fourth embodiment of the present disclosure wherein a front side of the FM layer in FIG. **8** is recessed to a WS throat height.

[0027] FIG. **10** is a down-track cross-sectional view of a fifth embodiment of the present disclosure wherein a second SP layer is inserted between the FM layer and MP trailing side in FIG. **9** so that spin polarized current from the FM layer flows through both of the first and second SP layers to the WS and MP, respectively.

[0028] FIG. **11** is a down-track cross-sectional view showing a sixth embodiment of the present disclosure that is a modification of the fifth embodiment wherein the lead to the FM layer is stitched.

[0029] FIG. **12** is a down-track cross-sectional view depicting a seventh embodiment of the present disclosure wherein the FM layer in FIG. **9** is omitted, and the SP layer has a first portion adjoining the MP that has a recessed front side while a second portion of the SP layer contacting the WS has a front side at the ABS.

[0030] FIGS. **13A**, **14A**, **16A-18A** and FIGS. **19-21** are ABS views that depict a sequence of steps in forming a SIAMR stack of layers between a MP trailing side and a WS according to an embodiment of the present disclosure.

[0031] FIGS. **13B**, **14B**, and **16B** are down-track cross-sectional views of the intermediate structures shown in FIGS. **13A**, **14A**, and **16A**, respectively, and FIG. **15** is a down-track cross-sectional view showing the patterning of a FM layer.

[0032] FIG. **17B** is a top-down view of the intermediate structure shown in FIG. **17A** that depicts a step of patterning a front portion of the SP layer according to an embodiment of the present disclosure.

[0033] FIG. **18B** is a top-down view of the intermediate structure in FIG. **18A** that depicts a step of forming a cross-track width in the SIAMR stack of layers according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0034] The present disclosure is a PMR writer with a SIAMR design wherein spin polarized current is injected into a FM layer and then across at least one SP layer in a WG before entering a main pole or a write shield in order to improve linear density, reliability, and ADC. In the drawings, the y-axis is in a cross-track direction, the z-axis is in a down-track direction, and the x-axis is in a direction orthogonal to the ABS and towards a back end of the writer structure. Thickness refers to a down-track distance, width is a cross-track distance, and height is a distance from the ABS in the x-axis direction. The terms “downward” and “upward” when referring to write field and return field directional arrows in the drawings indicate a direction that is into the ABS and toward a back end of the PMR writer for “upward”, and out of the ABS and toward a magnetic medium for “downward”. The terms “MP field” and “write field” may be used interchangeably. Although the exemplary embodiments depict a single PMR writer, the present disclosure anticipates that two or more PMR writers may be formed on a slider and each of the PMR writers comprises a SIAMR device described herein.

[0035] Referring to FIG. **1**, a head gimbal assembly (HGA) **100** includes a magnetic recording head **1** comprised of a slider and a PMR writer structure formed thereon, and a suspension **103** that elastically supports the magnetic recording head. The suspension has a plate spring-like load beam **222** formed with stainless steel, a flexure **104** provided at one end portion of the load beam, and a base plate **224** provided at the other end portion of the load beam. The slider portion of the magnetic recording head is joined to the flexure, which gives an appropriate degree of freedom to the magnetic recording head. A gimbal part (not shown) for maintaining a posture of the magnetic recording head at a steady level is provided in a portion of the flexure to which the slider is mounted.

[0036] HGA **100** is mounted on an arm **230** formed in the head arm assembly **103**. The arm moves the magnetic recording head **1** in the cross-track direction y of the magnetic recording medium **140**.

One end of the arm is mounted on base plate **224**. A coil **231** that is a portion of a voice coil motor is mounted on the other end of the arm. A bearing part **233** is provided in the intermediate portion of arm **230**. The arm is rotatably supported using a shaft **234** mounted to the bearing part **233**. The arm **230** and the voice coil motor that drives the arm configure an actuator.

[0037] Next, a side view of a head stack assembly (FIG. 2) and a plan view of a magnetic recording apparatus (FIG. 3) wherein the magnetic recording head **1** is incorporated are depicted. The head stack assembly **250** is a member to which a first HGA **100-1** and second HGA **100-2** are mounted to arms **230-1**, **230-2**, respectively, on carriage **251**. A HGA is mounted on each arm at intervals so as to be aligned in the perpendicular direction (orthogonal to magnetic medium **140**). The coil portion (**231** in FIG. 1) of the voice coil motor is mounted at the opposite side of each arm in carriage **251**. The voice coil motor has a permanent magnet **263** arranged at an opposite position across the coil **231**.

[0038] With reference to FIG. 3, the head stack assembly **250** is incorporated in a magnetic recording apparatus **260**. The magnetic recording apparatus has a plurality of magnetic media **140** mounted to spindle motor **261**. For every magnetic recording medium, there are two magnetic recording heads arranged opposite one another across the magnetic recording medium. The head stack assembly and actuator except for the magnetic recording heads **1** correspond to a positioning device, and support the magnetic recording heads, and position the magnetic recording heads relative to the magnetic recording medium. The magnetic recording heads are moved in a cross-track of the magnetic recording medium by the actuator. The magnetic recording head records information into the magnetic recording media with a PMR writer element (not shown) and reproduces the information recorded in the magnetic recording media by a magnetoresistive (MR) sensor element (not shown).

[0039] Referring to FIG. 4, magnetic recording head **1** comprises a combined read-write head. The down-track cross-sectional view is taken along a center plane (**44-44** in FIG. 5) formed orthogonal to the ABS **30-30**, and that bisects the main pole layer **14**. The read head is formed on a substrate **81** that may be comprised of AlTiC (alumina+TiC) with an overlying insulation layer **82** that is made of a dielectric material such as alumina. The substrate is typically part of a slider formed in an array of sliders on a wafer. After the combined read head/write head is fabricated, the wafer is sliced to form rows of sliders. Each row is typically lapped to afford an ABS before dicing to fabricate individual sliders that are used in a magnetic recording device. A bottom shield **84** is formed on insulation layer **82**.

[0040] A magnetoresistive (MR) element also known as MR sensor **86** is formed on bottom shield **84** at the ABS **30-30** and typically includes a plurality of layers (not shown) including a tunnel barrier formed between a pinned layer and a free layer where the free layer has a magnetization (not shown) that rotates in the presence of an applied magnetic field to a position that is parallel or antiparallel to the pinned layer magnetization. Insulation layer **85** adjoins the backside of the MR sensor, and insulation layer **83** contacts the backsides of the bottom shield and top shield **87**. The top shield is formed on the MR sensor. An insulation layer **88** and a top shield (S2B) layer **89** are sequentially formed on the top magnetic shield. Note that the S2B layer **89** may serve as a flux return path (RTP) in the write head portion of the combined read/write head. Thus, the portion of the combined read/write head structure formed below layer **89** in FIG. 4 is typically considered as the read head. In other embodiments (not shown), the read head may have a dual reader design with two MR sensors, or a multiple reader design with multiple MR sensors.

[0041] The present disclosure anticipates that various configurations of a write head may be employed with the read head portion. In the exemplary embodiment, magnetic flux known as write field **70** in main pole (MP) layer **14** is generated with flowing a write current (I_w) through bucking coil **80b** and driving coil **80d** that are below and above the MP layer, respectively, and are connected by interconnect **51**. Magnetic flux **70** exits the MP layer at pole tip **14p** at the ABS **30-30** and is used to write a plurality of bits on magnetic media **140**. Magnetic flux **70b** returns to the MP

through a trailing loop comprised of trailing shields **17**, **18**, PP3 shield **26**, and top yoke **18x**. There is also a leading return loop for magnetic flux **70a** that includes leading shield **11**, leading shield connector (LSC) **33**, S2 connector (S2C) **32**, return path **89**, and back gap connection (BGC) **62**. The magnetic core may also comprise a bottom yoke **35** below the MP layer. Dielectric layers **10**, **13**, **36-39**, and **47-49** are employed as insulation layers around magnetic and electrical components. A protection layer **27** covers the PP3 trailing shield and is made of an insulating material such as alumina. Above the protection layer and recessed a certain distance u from the ABS **30-30** is an optional cover layer **29** that is preferably comprised of a low coefficient of thermal expansion (CTE) material such as SiC. Overcoat layer **28** is formed as the uppermost layer in the write head. [0042] Referring to FIG. 5, an ABS view is depicted of the PMR writer in FIG. 4 according to an embodiment of the present disclosure. The main pole (MP) has a MP tip **14p** with track width w , trailing side **14t1**, leading side **14b1**, and two sides **14s** formed equidistant from a center plane **44-44** in an All Wrap Around (AWA) shield structure. There is a write gap **16** with thickness t on the MP trailing side, side gaps **15** adjoining each MP side, and a leading gap **13** below the MP leading side. Leading shield **11** contacts the leading gap and a bottom surface of side shields **12** at plane **42-42**. Side shields (SS) have an inner side **12s** adjoining a side gap, and far sides **60**, **61** at an outer side of the AWA shield structure. The trailing shield (TS) structure comprises a first trailing shield hereinafter referred to as the write shield (WS) **17** with a magnetic saturation value from 19 kiloGauss (kG) to 24 kG, and with a bottom surface **17b** on the WG except over a SIAMR stack of layers **2-6** each having a width $\leq \frac{1}{2} w$ on each side of the center plane. WS sides **17s** are formed coplanar with WG sides **16s** on each side of the center plane. The TS structure also includes a second TS **18** formed on the WS top surface and sides **17t** and **17s**, respectively, on WG sides **16s**, and on a top surface of the side shields **12** at plane **41-41**. Plane **41-41** includes the MP trailing side at the ABS. Plane **42-42** is parallel to plane **41-41**, and includes the MP leading side at the ABS. Each of the SS, leading shield, TS structure, and MP are comprised of one or more of CoFe, NiFe, and CoFeNi.

[0043] According to a first embodiment of the present disclosure, the SIAMR stack of layers has a first spin killing (SK) layer **2**, non-magnetic spacer **3**, second SK layer **4**, FM layer **5**, and spin preserving (SP) layer **6** sequentially formed on MP trailing side **14t1**. The SIAMR stack of layers has thickness t equivalent to that of WG layer **16** between plane **41-41** and WS **17**, and a width that is typically $\leq w$. Each SK layer is also known as a non-spin preserving layer and prevents spin polarized electrons from the FM layer from reaching the MP trailing side, and is preferably one of Ta, W, Pt, Ru, Ti, Ir, or Cr. The non-magnetic spacer is comprised of Ta, Ru, W, or Cr, and serves to improve the growth of the FM layer for better magnetic properties. The FM layer may be a single layer or multilayer that is comprised of one or more of Fe, Co, CoFe, NiFe, CoFeNi, and alloys thereof such as CoB, FeB, CoFeB, and CoFeNiB, or alloys with one or more of Ta, Zr, Re, and Mo. The SP layer is non-magnetic and comprises one or more of Cu, Au, Ag, Ru, Cr, and Al, and is responsible for having sufficient spin diffusion length to allow spin polarized electrons (not shown) to substantially remain in their original orientation while traversing from the FM layer to WS **17** (and to MP **14** in the fifth and sixth embodiments described later).

[0044] The first embodiment is a modification of the spin flipping element disclosed in related U.S. Pat. No. 10,325,618. We have discovered that by replacing a single SK layer with a stack comprised of a non-magnetic spacer **3** between two SK layers **2** and **4** in the SIAMR stack of layers, performance characteristics including electrical conductivity, thermal conductivity, electro-thermal robustness, and magnetic damping properties are simultaneously optimized.

[0045] Referring to FIG. 6A, a down-track cross-sectional view of the first embodiment is shown where the SIAMR stack of layers **2-6** is formed between MP **14** and WS **17**. First SK layer **2** forms a first interface that is also referred to as the MP/SK interface with MP trailing side **14t1**, and SP layer **6** forms a second interface also known as the SP/WS interface with WS bottom surface **17b**. MP local magnetization **14m** that is adjacent to the MP/SK interface is generally in the direction of

the write gap field flux (H.sub.WG), which is from the MP trailing side to WS bottom surface. There is also a local WS magnetization **17m** adjacent to the SP/WS interface that is substantially parallel to H.sub.WG. Moreover, FM layer **5** has magnetization **5m** substantially parallel to H.sub.WG. Other layers in the PMR writer are omitted in order to focus on the path of applied current I.sub.a while writing a transition where write field **70** is oriented downward at the ABS **30-30** and into a magnetic medium (not shown), and where return field **70b** is orthogonal to the ABS and upward into the WS. In the exemplary embodiment, each of the SIAMR stack of layers has a front side at the ABS, and a backside at a WS throat height h where the WS bottom surface connects with WS side **17s**. Thus, front sides **5f** and **6f** of the FM layer and SP layer, respectively, are coplanar, and FM layer backside **5e** and SP layer backside **6e** are at height h. Lead **57** carries I.sub.a into MP **14** from direct current (dc) source **50** while lead **58** carries I.sub.a back to the dc source from TS **17**. In other embodiments (not shown), the SIAMR stack of layers may have a backside at a height unequal to h.

[0046] A key feature is that when I_a flows from the MP **14** to WS **17**, spin polarized current from FM layer **5** traverses the SP layer **6** and enters the WS, and produces a magnetization (not shown) proximate to the SP/WS interface that enhances local WS magnetization **17m** and return field **70b**. I.sub.a current density is less than required to flip FM magnetization **5m** in STRAMR devices. Therefore, improved reliability is realized because of a reduced risk of electromigration within the SIAMR metal or alloy layers. During a transition (not shown) when the write field is orthogonal to the ABS **30-30** and upward into the MP, and the directions of I.sub.a, magnetizations **5m**, **14m**, and **17m**, and return field **70b** are reversed, there is no spin torque applied to local MP magnetization **14m** because SK layers **2** and **4** prevent spin polarized electrons from reaching the MP.

[0047] An alternative first embodiment is depicted in FIG. **6B** where the SIAMR layers are formed in reverse stacking order on MP trailing side **14t1** such that SP layer **6** adjoins the MP trailing side and forms a MP/SP interface while SK layer **2** contacts the WS bottom surface **17b** and forms a SK/WS interface. This reverse SIAMR configuration is advantageous for boosting write field **70** during a transition when the write field is oriented orthogonal to the ABS **30-30** and into the MP **14**, and when I.sub.a is applied from the WS to MP. Here, local MP magnetization **14m**, FM magnetization **5m**, local WS magnetization **17m**, and return field **70b** are also opposite to the direction shown in FIG. **6A**. In this case, spin polarized current from FM layer **5** traverses the SP layer and enters the MP to generate a magnetization adjacent to the MP trailing side that enhances the local MP magnetization **14m**, and write field **70** that is in an upward direction. When the write field is reversed and I.sub.a is applied from the MP to WS, there is no spin torque produced on WS magnetization **17m** because of the presence of SK layers **2**, **4** that prevent spin polarized electrons from reaching WS bottom surface **17b**.

[0048] Referring to FIG. **7**, a second embodiment of the present disclosure is depicted and is a modification of the first embodiment where SK layers **2** and **4**, and spacer **3** are replaced with WG layer **16** that is a dielectric material. Furthermore, current I.sub.a is injected through lead **57** and then into a contact **59** before entering the FM layer **5** since there is no longer a conductive pathway from MP **14** to the FM layer. The FM layer backside **5e** is now at a height substantially greater than height h₁ where the MP tapered trailing side **14t1** connects with MP top surface **14t2** at corner **14c**. Thus, the FM layer has a bottom surface **5b** that faces the MP and is separated from MP trailing side by WG layer **16**, and from MP top surface by insulation layer **47**. FM layer top surface **5t** adjoins contact **59** proximate to end **5e**. Otherwise, all aspects of the SP layer **6** are retained from the first embodiment in FIG. **6A** including a front side **6f** at the ABS **30-30**, and a backside **6e** at the WS TH, which is height h. Accordingly, spin polarized electrons from the FM layer traverse SP layer **6** and generate magnetization that enhances local WS magnetization **17m** proximate to the SP/WS interface, and return field **70b** when write field **70** is out of the MP and orthogonal to the ABS **30-30**, and the return field is in an upward direction. Again, linear density and ADC are improved. When writing a transition where the write field, return field, I.sub.a, and magnetizations

14m, **5m**, and **17m** are reversed, there is no enhancement to the write field because the WG layer prevents spin polarized electrons from reaching the MP trailing side.

[0049] Referring to FIG. 8, a third embodiment of the present disclosure is illustrated and is a modification of the second embodiment in FIG. 7 where SP layer backside **6e** is extended farther from the ABS **30-30** to a height **h2** where $h2 > h$. The additional height of the SP layer **6** between front side **6f** and backside **6e** is expected to allow more spin polarized current from the FM layer **5** to the SP/WS interface thereby generating greater magnetization to further enhance WS magnetization **17m** and write field **70b** (with the same current density in FM layer **5**) compared with the SIAMR structure in FIG. 7. Optionally, less current density (same total current) in the FM layer produces an equal amount of magnetization to enhance WS magnetization and return field compared with previous embodiments to improve device reliability. Otherwise, all features and advantages of the second embodiment are retained in the third embodiment.

[0050] In FIG. 9, a fourth embodiment of the present disclosure is shown where the SIAMR structure in FIG. 8 is modified by recessing FM layer front side **5f** to height **h**. Thus, WG layer **16** fills the entire space between the ABS **30-30** and FM layer front side, and SP layer **6** is the only SIAMR layer exposed at the ABS and in contact with WS **17**. This design is expected to allow lower current density between the FM layer **5** and SP layer since this scheme allows a greater overlap of the aforementioned layers in the cross-track direction in embodiments where there is a larger WG cavity (not shown) at a height greater than **h** than between the ABS and the WS TH at height **h**. As a result, there is a significant improvement in reliability at the same performance compared with earlier embodiments where the write field **70** is in a downward direction out of the ABS, the return field **70b** is upward from the ABS, and **la** is from the FM layer and across the SP layer into the WS **17** to enhance local WS magnetization **17m** and the return field.

[0051] Referring to FIG. 10, a fifth embodiment of the present disclosure is depicted and is a modification of the embodiment in FIG. 9 where a portion of the WG layer **16** adjoining MP trailing side **14t1** is replaced with a first SP layer (SP1) **6a** having a backside **6e1** at height **h2**, and a front side **6f1** at the ABS **30-30**. Thus, the SP1 forms a MP/SP1 interface with the MP trailing side, and adjoins a portion of FM layer bottom surface **5b**. The second SP layer (SP2) **6b** (former SP layer **6** in FIG. 9) contacts a portion of FM layer top surface **5t**, and forms a SP2/WS interface with WS bottom surface **17b**. The SP2 has a front side **6f2** at the ABS, and a backside **6e2** at height **h2**. There is a first return lead **58a** between TS **17** and dc source **50** that carries current $I_{sub.b}$, and a second return lead **58b** between MP **14** and the dc source that carries $I_{sub.a}$. In this embodiment, lead **57** is connected to contact **59**, which injects $(I_{sub.a} + I_{sub.b})$ from the dc source to FM layer **5**. In alternative embodiments (not shown) where the FM layer front side **5f** is moved forward to the ABS, one or both of SP1 and SP2 backsides **6e1** and **6e2**, respectively, may also be moved forward to height **h**, or maintained at height **h2**.

[0052] An advantage of the SIAMR layout in the fifth embodiment compared with previous embodiments is that write field **70** and return field **70b** are simultaneously enhanced as a result of spin polarized current from FM layer **5** through SP1 **6a** and into MP tip **14p** to generate magnetization (not shown) that enhances local MP magnetization **14m** and the write field, and from the FM layer through SP2 **6b** and into WS **17** to produce magnetization that increases local WS magnetization **17m** and the return field.

[0053] Note that the fifth embodiment provides similar benefits when the direction of the write field **70**, return field **70b**, and magnetizations **5m**, **14m**, and **17m** are reversed (not shown) because $I_{sub.a}$ and $I_{sub.b}$ are maintained in the same direction. Thus, spin polarized electrons traversing through SP1 **6a** from FM layer **5** will continue to enhance local MP magnetization **14** and the write field, and spin polarized current through SP2 **6b** will continue to increase local WS magnetization **17m** and the return field.

[0054] Referring to FIG. 11, a sixth embodiment of the present disclosure is shown that is a modification of the fifth embodiment where SP1 and SP2 front sides **6f1** and **6f2**, respectively, as

well as FM layer front side **5f** are at the ABS **30-30**, and where SP1 and SP2 backsides **6e1** and **6e2**, respectively, are at height **h2**. A key feature is that lead **57** from the dc source **50** to FM layer **5** in FIG. **10** is now stitched and comprised of a SP material or spin killing material in stitched layer **7** having one end at contact **59**, and an opposite end that overlaps a top surface **5t** of the FM layer proximate to end **5e**. Accordingly, a portion of stitched layer bottom surface **7b** adjoins the FM layer top surface. The stitched lead is advantageously employed to reduce the lead resistance and thereby lower temperature in the SIAMR device to improve reliability. It should be understood that the lead modification described for the sixth embodiment may be applied to any of the second through fifth embodiments with similar benefits. In alternative embodiments (not shown) where the FM layer has a front side at the ABS, one or both of SP1 **6a** and SP2 **6b** may have a backside at height **h**.

[0055] According to a seventh embodiment shown in FIG. **12**, the SIAMR device in FIG. **7** is modified with the removal of FM layer **5** such that a single SP layer extends from the MP trailing side **14t1** to the WS bottom surface **17b**. Thus, a first portion **6a** of the SP layer has a recessed front side **6f1** at height **r** that may be proximate to the WS TH, a backside **6e** at height **h2**, and forms a MP/SP interface with MP trailing side. Preferably, height **r** is from 1 nm to 20 nm and slightly larger than the effective throat height on the trailing shield, and **h2** can be from 10 to 200 nm larger than **r** from the ABS **30-30**. In the exemplary embodiment, the MP trailing side connects with MP top surface **14t2** at corner **14c** that is at height **h1** from the ABS where **h1**>**h2**. A second portion **6b** of the SP layer has backside **6e** at height **h2**, a front side **6f2** at the ABS, adjoins first portion **6a** between height **r** and height **h2**, and forms a SP/WS interface with WS bottom surface **17b**. The first portion of SP layer is recessed from the ABS so that local MP magnetization **14m** proximate to the ABS is not reduced. Note that lead **57** now connects dc source **50** with MP **14**. Here, the MP proximate to the MP trailing side serves as the FM layer which spin polarizes I.sub.a that flows from the MP and across the SP layer and into the WS **17**. Similar to previous embodiments, the spin polarized electrons produce a magnetization proximate to the WS bottom surface that enhances local WS magnetization **17m** and return field **70b**.

[0056] For all embodiments, the advantage of injecting spin polarized electrons from the FM layer across the SP layer and into one or both of the MP and WS generates magnetization at the MP/SP interface and SP/WS interface, respectively, that increases local magnetization at one or both of the interfaces and enhances one or both of the write field and return field depending on the direction of the current flow (except in the fifth and sixth embodiments). As a result, the PMR writer has improved performance based on greater linear density and higher ADC. Furthermore, a PMR writer having a SIAMR structure in the WG is expected to exhibit greater reliability because a lower current density for I.sub.a is required than in PMR writers employing a STRAMR device in the WG.

[0057] A process sequence for fabricating a magnetic flux guiding device in a WG and with a front side at the ABS has been described in related U.S. Pat. No. 10,424,326. A similar sequence of steps may be used to form a SIAMR stack of layers having a width **w** in a WG, and a backside at a WS TH as shown in the first embodiment.

[0058] A process sequence for fabricating the SIAMR device of the second through fourth embodiments is illustrated starting at FIG. **13A**. The partially formed writer structure including MP tip **14p** that adjoins side gaps **15** and leading gap **13** is provided according to a conventional process sequence. Side shield top surfaces **12t** are coplanar with a trailing edge of the MP tapered trailing side **14t1** at plane **41-41**, which is slightly offset from being orthogonal to the subsequently formed ABS plane. FIG. **13B** shows the down-track cross-sectional view at plane **44-44** in FIG. **13A**. MP tapered trailing side **14t1** has a taper angle δ (generally greater than 0 degrees) and is coplanar with a tapered front side **47f** of insulation layer **47** made of Al.sub.2O.sub.3 or SiO.sub.2 that is formed on MP top surface **14t2**. Tapered front side connects with dielectric layer top surface **47t** that is above and essentially parallel to the MP top surface. Note that the eventual ABS,

hereinafter referred to as ABS plane **30-30**, is not determined until a lapping process is performed after all layers in the PMR writer structure are formed.

[0059] Referring to FIG. **14A**, the next step in the fabrication process is sequential deposition of WG layer **16** and FM layer **5** on MP tapered trailing side **14t1**. FIG. **14B** shows that the WG layer and FM layer continue toward a back end of the writer structure and are also formed on insulation layer **47**.

[0060] As shown in FIG. **15**, a first photoresist layer is coated on FM layer **5**, and then exposed with the use of a chrome on quartz mask, for example, and developed with a photolithography process to form photoresist mask **52** having a front side **52f** that is recessed to height h from the ABS plane **30-30** and thereby forming an opening **53a** that exposes a portion of FM layer top surface **5t**. Note that opening **53a** may be omitted when forming a FM layer with a front side at the ABS as in the second and third embodiments. A second opening **53b** is also formed during the photoresist patterning process. As a result, photoresist mask backside **52e** is formed above MP top surface **14t2**, and will be used to define a FM layer backside in a subsequent step.

[0061] In FIG. **16A**, an ABS view is shown after a reactive ion etch (RIE) or ion beam etch (IBE) process is employed to transfer openings **53a**, **53b** through the FM layer **5** and stop on WG layer top surface **16t**. Thereafter, photoresist mask **52** is removed by a conventional method. FIG. **16B** depicts a down-track cross-sectional view of the intermediate structure in FIG. **16A**. The FM layer now has a front side **5f** that is recessed to height h from the ABS plane **30-30**, and a backside **5e** above MP top surface **14t2**.

[0062] Referring to FIG. **17A**, an additional WG layer **16a** is optionally deposited on exposed portions of WG layer **16** between FM layer front side **5f** and the ABS plane, and behind FM layer backside **5e**. In the exemplary embodiment, the WG layers **16**, **16a** are made of the same dielectric material and shown hereinafter as a single WG layer **16**. Then, SP layer **6** is deposited on the FM layer (not shown) and on the WG layer. A sputter deposition or ion beam deposition tool may be used to deposit each SIAMR layer including the FM layer **5** and SP layer. A second photoresist is coated and then exposed with the aid of another mask and developed by a photolithography process to yield photoresist mask **54** having a full width w_1 between sides **54s** that are coplanar with sides **60**, **61** of the AWA shield structure described previously.

[0063] FIG. **17B** shows a top-down view of the intermediate structure in FIG. **17A** where the underlying MP has sides indicated by the dashed line **14s**. Dashed line **14x** is between corners **14c** and represents the interface where the MP tip portion **14p** adjacent to the MP trailing side **14t1** meets the back portion of MP **14** below MP top surface **14t2**. Corner **14c** is also shown in a down-track cross-sectional view in FIG. **9**. Returning to FIG. **17B**, backside **54b** of the second photoresist mask **54** is at height h_2 where $h_2 > h$ and will be used in a later step to define a backside of SP layer **6**. Photoresist mask front side **54f** may be at the ABS plane **30-30** or on an opposite side (not shown) of the ABS plane with respect to backside **54b**.

[0064] FIG. **18A** depicts the intermediate structure in FIG. **17A** after a RIE or IBE step is performed to transfer the shape in second photoresist mask **54** through exposed portions of SP layer **6** and stopping on FM layer **5** thereby forming a SP layer backside **6e** at height h_2 (FIG. **18B**). The resulting opening (not shown) behind the SP layer is refilled with WG layer **16**. The second photoresist mask is removed. Next, a third photoresist is coated and then exposed with a new mask and developed to form a third photoresist mask **56** having width w and sides **56s** that are equidistant from center plane **44-44**. Openings **55** are formed that expose portions of SP layer top surface **6t1** at distances greater than $\frac{1}{2} w$ from the center plane. Photoresist mask front side **56f** may be on an opposite side of plane **30-30** than MP sides **14s** since the ABS is not determined until a lapping process at the end of the fabrication sequence.

[0065] Referring to FIG. **19**, a RIE or IBE process is performed to transfer openings **55** through exposed portions of WG layer **16**, and FM layer **5** before stopping on SS top surface **12t** and side gap **15** on each side of center plane **44-44**. As a result, SP layer **6** has a constant width w from plane

30-30 to a backside (not shown) at height h_2 , and FM layer **5** has width w from front side that is at height h to a backside **5e** described earlier. Thereafter, WG layer **16** is deposited on SS top surface and the side gap to thickness t . The third photoresist mask is removed with a conventional method. [0066] Referring to FIG. **20**, WS **17** is deposited on WG layer **16** and on SP layer **6**. Next, a fourth photoresist layer is coated on the WS and is exposed with a fourth mask and developed to generate a fourth photoresist mask **58** having width w_2 where $w_2 > w$, and sides **58s** that are equidistant from center plane **44-44**. Openings **57** adjacent to sides **58s** expose a portion of WS top surface **17t**. [0067] FIG. **21** depicts the intermediate structure in FIG. **20** after a RIE or IBE is performed to transfer openings **57** through WS **17** and WG layer **16**, and stopping on SS top surface **12t**. After the fourth photoresist mask is removed with a conventional process, the second TS **18** is plated on SS top surface **12t**, WS top surface **17t**, and adjoins WG sides **16s** and WS **17s**. Thereafter, a well known sequence of steps is followed to form overlying layers in the PMR writer. Finally, a lapping step is used to form the ABS at plane **30-30** described previously. [0068] While the present disclosure has been particularly shown and described with reference to, the preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made without departing from the spirit and scope of this disclosure.

Claims

1. A spin injection assisted magnetic recording (SIAMR) structure, comprising: (a) a main pole (MP) that is configured to generate a magnetic (write) field which is directed orthogonal to an air bearing surface (ABS) and through a MP tip at the ABS; (b) a write shield (WS) with a side at the ABS through which a return magnetic field passes orthogonal to the ABS, and having a bottom surface that faces a MP trailing side and a throat height (WS TH) at a first height from the ABS; and (c) a SIAMR stack of layers formed in a write gap (WG) and comprising: (1) a spin preserving (SP) layer having a front side at the ABS and contacting the WS bottom surface, a back side at a height greater than the WS TH, and wherein the SP layer adjoins a first side of a ferromagnetic (FM) layer and is configured to conduct spin polarized current from the FM layer into the WS wherein the spin polarized current produces a magnetization proximate to the WS bottom surface that enhances a local WS magnetization and the return magnetic field; and (2) the FM layer having a magnetization substantially in a direction of a WG field flux between the MP and WS, the WG separates a front side of the FM layer from the ABS, and separates the FM layer from the MP trailing side, and wherein the FM layer is connected through a first lead to a direct (dc) current source, and the WS is connected to the dc source through a second lead such that when a current (I_a) is injected into the FM layer, the FM layer spin polarizes current (I_a) that flows through the SP layer and returns to the dc source through the second lead.
2. The SIAMR structure of claim 1 wherein the SP layer has a backside at a height greater than the WS TH.
3. The SIAMR structure of claim 1 wherein the SP layer is one or more of Cu, Au, Ag, Ru, Cr, and Al.
4. The SIAMR structure of claim 1 wherein the FM layer is a single layer or multilayer comprised of one or more of Fe, Co, CoFe, NiFe, CoFeNi, and alloys thereof including CoB, FeB, CoFeB, and CoFeNiB, or alloys with one or more of Ta, Zr, Re, and Mo.
5. The SIAMR structure of claim 1 wherein the front side of the FM layer is recessed to a height that is equal to the WS TH.
6. A head gimbal assembly (HGA), comprising: (a) a magnetic recording head comprising the SIAMR structure of claim 1; and (b) a suspension that elastically supports the magnetic recording head, wherein the suspension has a flexure to which the magnetic recording head is joined, a load beam with one end connected to the flexure, and a base plate connected to the other end of the load

beam.

7. A magnetic recording apparatus, comprising: (a) the HGA of claim **6**; (b) a magnetic recording medium positioned opposite to the slider; (c) a spindle motor that rotates and drives the magnetic recording medium; and (d) a device that supports the slider, and that positions the slider relative to the magnetic recording medium.

8. A spin injection assisted magnetic recording (SIAMR) structure, comprising: (a) a main pole (MP) that is configured to generate a magnetic (write) field which is directed orthogonal to an air bearing surface (ABS) and through a MP tip at the ABS; (b) a write shield (WS) with a side at the ABS through which a return magnetic field passes orthogonal to the ABS, and having a bottom surface that faces a MP trailing side; and (c) a spin preserving (SP) layer having a first portion adjoining the MP trailing side and a second portion contacting the WS bottom surface, and wherein the first and second portions interface with each other and are configured to allow spin polarized current from the MP trailing side to enter the WS and thereby generate a magnetization proximate to the WS bottom surface that enhances a local WS magnetization and the return magnetic field, or allow spin polarized current from the WS bottom surface to enter the MP and generate a magnetization proximate to the MP trailing side that enhances a local MP magnetization and the write field.

9. The SIAMR structure of claim 8 wherein a front side of the SP layer first portion is recessed from the ABS, and wherein the second portion of the SP layer has a front side at the ABS.

10. The SIAMR structure of claim 8 wherein each of the first and second portions has a backside at a first height that is greater than a WS throat height.

11. The SIAMR structure of claim 8 wherein the SP layer is one or more of Cu, Au, Ag, Ru, Cr, and Al.

12. The SIAMR structure of claim 8 wherein the recessed front side of the SP layer first portion is from 1 nm to 20 nm from the ABS.

13. The SIAMR structure of claim 8 wherein a dielectric (WG) layer is formed between the ABS and the recessed front side of the SP layer first portion.

14. A head gimbal assembly (HGA), comprising: (a) a magnetic recording head comprising the SIAMR structure of claim 8; and (b) a suspension that elastically supports the magnetic recording head, wherein the suspension has a flexure to which the magnetic recording head is joined, a load beam with one end connected to the flexure, and a base plate connected to the other end of the load beam.

15. A magnetic recording apparatus, comprising: (a) the HGA of claim **14**; (b) a magnetic recording medium positioned opposite to the slider; (c) a spindle motor that rotates and drives the magnetic recording medium; and (d) a device that supports the slider, and that positions the slider relative to the magnetic recording medium.

16. A method of forming a spin injection assisted magnetic recording (SIAMR) structure, comprising: (a) providing a main pole (MP) with a tapered trailing side that extends from an air bearing surface (ABS) plane to a height (h_1) where the MP tapered trailing side connects with a MP top surface aligned orthogonal to the ABS plane and having a first dielectric layer formed thereon, and wherein the MP is separated from a side shield on each side of a center plane by a side gap, and the MP trailing side has a track width that is bisected by the center plane; (b) sequentially depositing a first write gap (WG) layer and a ferromagnetic (FM) layer on the MP tapered trailing side, first dielectric layer, side gaps, and side shields, and forming an opening to expose a front side of the FM layer that is separated from a first plane by a first height (h), and exposes a back side of the FM layer; (c) sequentially depositing a second WG layer to fill the opening, and then a spin preserving (SP) layer on the first and second WG layers and on the FM layer, wherein the first and second WG layers, SP layer, and FM layer form a SIAMR stack of layers; (d) forming a backside on the SP layer that is a second height (h_2) from the first plane where $h_2 > h$; (e) patterning the SIAMR stack of layers such that each layer therein has a first cross-track width (w) that is bisected

by the center plane, and wherein the patterning exposes a top surface of the side shields and side gaps; (f) sequentially depositing a third WG layer to form a thickness (t) on the top surfaces of the side shields and side gaps, and a write shield (WS) on the third WG layer and on the SP layer; and (g) performing a lapping process to form an air bearing surface (ABS) at the first plane.

17. The method of claim 16 wherein the FM layer is a single layer or multilayer comprised of one or more of Fe, Co, CoFe, NiFe, CoFeNi, and alloys thereof including CoB, FeB, CoFeB, and CoFeNiB, or alloys with one or more of Ta, Zr, Re, and Mo.

18. The method of claim 16 wherein the SP layer is one or more of Cu, Au, Ag, Ru, Cr, and Al.

19. The method of claim 16 wherein the backside of the FM layer is at a height greater than h2 from the ABS.

20. The method of claim 16 wherein h2 is less than h1.
